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Addition of air-nanobubble water to mitigate the inhibition of high salinity on co-production of hydrogen and methane from two-stage anaerobic digestion of food waste

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ABSTRACT

Most previous studies have focused on the effects of salt on anaerobic digestion (AD) for hydrogen or methane production. However, the effects of salt on two-stage AD and the related approaches to mitigate the adverse effects of high salinity on hydrogen and/or methane production were seldom addressed. In this study, addition of Air-nanobubble water (Air-NBW) was adopted to mitigate the inhibition of high salinity on co-production of hydrogen and methane from two-stage AD of food waste (FW). In the Air-NBW added reactors with 0–30 g NaCl/L, hydrogen yield was increased by 21–65% with the subsequent methane yield elevated by 14–43% when compared to the corresponding deionized water (DW) group. This study for the first time confirmed that when two-stage AD of FW was exposed to the same salinity level, addition of Air-NBW could enhance enzymatic activities at the individual stage. Results of electron transport system (ETS) activity further demonstrate that addition of Air-NBW may promote the electron transfer associated with the synthesis of hydrogen and methane. Therefore, an efficient approach for hydrogen and methane recovery from the two-stage AD of FW under high salinity was proposed through improving microbial electron transfer and corresponding enzymatic activities at each stage via Air-NBW addition.

1. Introduction

Food and Agriculture Organization (FAO) has reported that about 1/3 of the food produced globally is lost or wasted in agroindustry and households (FAO, 2013a,b). As known, food waste (FW) generated by household consumption (household preparation and cooking) accounts for a large part of the total FW production, which was estimated as 7.7–10.14 million tonnes in Japan, with 94% being disposed by landfilling or incineration (Liu et al., 2016). However, landfilling is considered the least environmentally friendly option (FAO, 2013a,b). Yano and Sakai (2016) reported that Japan has historically adopted incineration as the primary treatment method for FW due to the lack of landfilling space, and currently the incinerator facilities are facing a seriously aging issue. In contrast, anaerobic digestion (AD) is considered an environmentally friendly technology for the management of FW because it can reduce negative impacts on the environment and recover bioenergy (H_2 and CH_4) and biofertilizers simultaneously.

Generally, the salinity of FW is high because salt (NaCl) is one of the seasonings widely used in the cooking process (He et al., 2019). Particularly in Japan, the salt intake is higher ($\geq$ 11 g per day) than in western societies according to the diverse eating habits (Kuwabara et al., 2020), indicating its higher salt concentration feature in FW. Nevertheless, the high salinity could cause an inhibitory effect on microorganisms' growth due to the increased osmotic pressure or dehydration effect (Tsapekos et al., 2019). Many previous studies have investigated the inhibition of salt on AD processes, including hydrolysis, acidogenesis, and methanogenesis. For instance, Lee et al. (2012) investigated the sodium inhibition on hydrogen production from a pure culture of *C. butyricum*, in which hydrogen production started to decrease at $\geq$ 2 g Na^+/L and the critical Na^+ concentration for ceased H_2 production was estimated as 32 g Na^+/L. Also, a sharp decrease in methane yield was noticed when NaCl was > 8 g/L (Anwar et al., 2016). Restated, under high salinity conditions, the high osmotic pressure and inappropriate enzyme linkages may lead to plasmolysis and loss of cell activity, which are the main reasons for inhibited AD processes (Zhang et al., 2020). Microorganisms indeed play a key role in the success of AD systems, and their activities largely influence the performance of AD systems. Liu et al. (2019) reported about 20% and 57% inhibition on the

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methanogenic activity at 5 and 10 g Na^+/L, respectively. Therefore, enhancement of microbial activity may be an appropriate strategy to mitigate salt inhibition on AD of high salinity FW.

Up to the present, most studies mainly focused on the effects of salt on AD for hydrogen or methane production, and few studies have been conducted to improve the efficiency of AD under high salt concentrations. Currently, few technologies are available to effectively mitigate the inhibition of high salinity on AD of FW, except for simple dilution with water or mixing with other low-salt organic wastes. Liu et al. (2019) attempted to ameliorate salt inhibition on methane production from the kitchen waste AD system by adding osmoprotectant (glycine betaine (GB)), and 2.0 and 2.5 g/L GB were found to be the optimal dosages for 5 and 10 g Na^+/L reactors with the resultant 29% and 62% increase in methane yield, respectively. However, the production cost of osmoprotectant is high, and a superfluous addition of osmoprotectant may adversely affect methane production.

Nanobubble water (NBW), containing tiny bubbles with a diameter < 1000 nm, has been reported to enhance the activities of microorganisms under both anaerobic and aerobic conditions for wastewater treatment (Agarwal et al., 2011; Temesgen et al., 2017). Additionally, NBW has been specifically applied to enhance the bioactivities of enzymes involved in AD systems of various organic wastes. For instance, Yang et al. (2019) and Wang et al. (2020a) found that addition of N_2- and O_2-NBW enhanced hydrolase activities in the AD of waste activated sludge and cellulase activity and electron transport system (ETS) activity in the AD of cellulose. Most recently, addition of Air-NBW was observed to not only enhance the hydrolase activities at the end of the hydrolysis/acidification stage but also improve the coenzyme F_{420} activity at the end of the methanogenesis stage in the two-stage AD of FW (Hou et al., 2021). However, the effects of NBW on microbial activities under high salinity conditions have not been determined. In addition, it is generally required to premix with some dilution water in a wet AD system to evenly distribute intermediate metabolites and microorganisms in the whole reactor. Besides, high total solids (TS) would lead to the accumulation of more toxic substances like free ammonia nitrogen, especially in the AD systems of FW, which may require water addition to adjust TS concentration appropriately (Wang et al., 2020b; Zhou et al., 2019). As such, NBW as a zero-pollution strategy may be considered as alternative water in the wet two-stage AD of FW, achieving enhanced microbial activity and thus mitigating salt inhibition on AD.

This study for the first time investigated the response behaviors of hydrogen and methane co-production from the two-stage AD of FW under different salt concentrations. Then, the effects of Air-NBW introduction on mitigating high salinity inhibition on AD of FW were evaluated by analyzing the variations of volatile fatty acids (VFAs), proteins, and carbohydrates. Ultimately, underlying mechanisms were preliminary explored by monitoring the enzymatic activities and ETS activity under Air-NBW addition at different salt concentrations. This study may provide an insight into the mechanisms involved in the two-stage AD of high salinity FW, targeting highly efficient energy recovery and sustainable management of FW.

2. Materials and methods

2.1. FW, inoculum and Air-NBW

According to Liu et al. (2017) and He et al. (2019), the simulated FW was comprised of rice (25%), noodle (22%), pork (16%), cabbage (29%), and tofu (8%). Before starting the experiments, the FW was smashed and homogenized into slurries using a food grinder and stored at −20°C. Inoculum applied in this study was anaerobically digested sludge sampled from the anaerobic digester of a wastewater treatment plant in Tsuchiura, Ibaraki Prefecture, Japan. The physicochemical characteristics of the FW and inoculum are summarized in Table 1.

Air-NBW was selected in this study, which was generated according to the procedures provided by Wang et al. (2019) and Yang et al. (2019).

Table 1
Physicochemical characteristics of the food waste and inoculum.

Parameter	Food waste	Inoculum
Total solids (TS, %)[a]	23.95 ± 0.37	0.80 ± 0.02
Volatile solid (VS, %)[a]	23.55 ± 0.32	0.55 ± 0.01
Total volatile fatty acids (VFAs, mg/L)	925.51 ± 21.72	9.77 ± 1.70
Soluble proteins (mg/L)	11226.51 ± 85.24	208.86 ± 17.78
Soluble carbohydrates (mg/L)	21836.00 ± 28.85	18.17 ± 0.26

[a] Wet weight basis.

The gas-water circulation method was adopted by using a NBW generator (HACK FB11, Japan), with the pressure and the flow rates of gas and water being controlled at 0.22 – 0.25 MPa, 0.05 L/min, and 3 L/min for 20 min. Air used for the Air-NBW generation was taken directly from the natural environment of the laboratory.

2.2. Effects of Air-NBW addition on the two-stage AD of FW at different salt concentrations

2.2.1. Hydrogen production (the first stage of AD)

All experiments of both stages were performed in the closed Schott Duran serum bottles of 250 mL with an effective volume of 150 mL each. In the first stage, according to the volatile solids (VS) content of FW and inoculum in Table 1, 6.5 g raw FW (wet weight) and 70 g inoculum were added to each reactor at a substrate to inoculum ratio (S/I) of 4:1 (VS basis). Then each bottle was diluted to the effective volume with deionized water (DW) or Air-NBW. As a result, the initial total VS was 12.76 ± 0.17 g/L. Afterwards, different amounts of salt (NaCl) were added to these reactors to obtain the pre-designed NaCl concentrations of 0, 5, 10, 20, and 30 g/L, respectively. At this stage, the initial pH was adjusted to the optimal pH (5.50 ± 0.20) considering the activity of hydrogenase. The reactors were denoted as FW + DW-0, FW + DW-5, FW + DW-10, FW + DW-20, and FW + DW-30 when DW was applied at salt concentrations of 0, 5, 10, 20, and 30 g/L, respectively. Similarly, the reactors were labelled as FW + Air-NBW-0, FW + Air-NBW-5, FW + Air-NBW-10, FW + Air-NBW-20, and FW + Air-NBW-30, respectively when Air-NBW was supplemented.

2.2.2. Methane production (the second stage of AD)

In the second stage, anaerobically digested sludge was firstly activated for about two weeks under 38 ± 1°C until no gas production, with TS and VS of 0.80 ± 0.01% and 0.55 ± 0.01%. 50 g remaining acidified slurry from the first stage was added as the substrate to each bottle together with 50 g activated inoculum (S/I = 2:1 (VS basis)) to produce methane. Then each bottle was diluted to the effective volume with DW or Air-NBW, resulting in an initial total VS of 6.01 ± 0.70 g/L with the initial pH being adjusted to 7.50 ± 0.10. Similarly, different amounts of NaCl were supplemented to each reactor to make the final NaCl concentrations to be 0, 5, 10, 20, or 30 g/L in the reactor with the same designations as in Section 2.2.1.

Before starting the experiments, all the reactors were flushed with high purity nitrogen (purity ≥ 99.9995%) around 2 min for 3 times to remove oxygen and then placed in a thermostatic incubator at 38 ± 1°C for 92 h in the first stage and 36 d in the second stage, respectively.

2.3. Analytical methods

2.3.1. General parameters and calculations

According to the standard methods (APHA, 2012), TS was analyzed by drying the samples at 105 ± 1°C till constant weight, and VS was recorded as the loss of dry mass after burning at 600 ± 2°C for 3 h. All liquid samples were centrifuged at 9,900 rpm for 20 min to obtain the supernatant and then filtered through 0.45 μm membrane filters before being analyzed. The concentrations of soluble proteins and carbohydrates were quantified using Lowry's method and Phenol-sulfuric acid

method with bovine serum albumin (BSA) and glucose (+) as standard, respectively (DuBois et al., 1956; Lowry, 1951). pH value was measured with a semi-solid pH meter (Testo 206, Germany). The concentrations of VFAs were determined by the gas chromatograph (GC-8A, Shimadzu, Japan) equipped with Unisole F-200 30/60 column and flame ionization detector.

During the entire AD process, the volume of biogas production was determined periodically by releasing the pressure in the serum bottle using gas-tight syringes of 50 and 20 mL to equilibrate with the room pressure (25°C, 1 atm). Afterwards, 1 mL biogas was sampled for the determination of H_2, CH_4, and CO_2 contents by using another gas chromatograph (GC-8A, Shimadzu, Japan).

The modified Gompertz model was used to fit the experimental data of each stage at different salt concentrations with or without Air-NBW addition as described previously (Hou et al., 2021). The inhibition degree was calculated according to Eq. (1).

$$\text{Inhibition degree } (\%) = \frac{\left(Y_{t,\ FW+DW,\ 0} - Y_{t,\ FW+DW/Air-NBW,\ i}\right)}{Y_{t,\ FW+DW,\ 0}} \times 100\% \quad (1)$$

where $Y_{t,\ FW+DW,\ 0}$ is the cumulative H_2 or CH_4 yield in FW + DW-0; $Y_{t,\ FW+DW/Air-NBW,\ i}$ is the cumulative H_2 or CH_4 yield in FW + DW or FW + Air-NBW at NaCl concentration of i (i = 5, 10, 20, or 30 g/L).

The undissociated acetic acid (HAc) and butyric acid (HBu) concentrations in all the reactors were calculated according to Eq. (2) (Cao and Zhao, 2009).

$$pH = pKa + \lg\frac{[A^-]}{[HA]} \quad (2)$$

where pKa is the dissociation constant (4.76 for HAc and 4.81 for HBu); $[A^-]$ and $[HA]$ are the dissociated acid and undissociated acid concentrations, respectively.

2.3.2. Microbial enzyme activities and ETS activity at each stage

The inhibition on hydrogen or methane production from the AD of FW under high salinity to a great extent depends on its impacts on microbial activities (Chen et al., 2018). More specifically, the main functional microorganisms in each stage are different in the two-stage AD process. Besides, the mechanisms involved in the influence of salinity on hydrolase activity in the first stage and coenzyme F_{420} in the second stage remain unclear. Therefore, it is necessary to evaluate the effect of high salinity on the major microbial enzyme activities and whether addition of Air-NBW could alleviate the inhibition of salt on microorganisms. In this study, the activities of α-glucosidase and protease at the end of the first stage and coenzyme F_{420} at the end of the second stage were measured according to the methods as previously described (Hou et al., 2021). For the determination of α-glucosidase, 0.1% p-nitrophenyl-α-D-glucopyranoside was used as substrate: 1 mL sample and 1 mL substrate were added in a 10 mL centrifuge tube, then 2 mL HAc-NaAc buffer (0.2 M, pH 7.5) was added to mix uniformly. After standing for 30 min at 55°C, 4 mL Na_2CO_3 (0.4 M) was added to terminate the reaction, and the absorbance was measured at 410 nm. Besides, 1% casein was used as the substrate to determine the activity of protease: 1 mL sample and 1 mL substrate were firstly added in a 10 mL centrifuge tube which was then placed at 55°C for 10 min. After 2 mL trichloroacetic acid (0.4 M) being added to terminate the reaction, centrifugation was performed and then filtered through 0.45 μm membrane filter. In the second step, 1 mL filtrate, 5 mL Na_2CO_3 (0.4 M), and 1 mL Folin-phenol were added successively, and the absorbance was measured at 640 nm after being placed at 40°C for 20 min.

In addition, the microbial activities during the hydrogen and methane production stages were assessed by monitoring the electron transport rate. The ETS activity was tested using the 2-(p-iodophenyl)-3-(p-nitrophenyl)-5-phenyltetrazolium chloride-electron transport system (INT-ETS) method, which has been described in detail previously (Wang et al., 2020a).

2.4. Statistical analysis

All the reactors operation were conducted in duplicate with the measurements being in triplicate. All the data were processed by Microsoft Excel (2010). One-way analysis of variance (ANOVA) was conducted to analyze the statistical difference between the conditions with or without the Air-NBW addition in the two-stage AD of FW at different salt concentrations. Statistical significance was assumed at $p < 0.05$.

3. Results and discussion

3.1. Effect of Air-NBW addition on hydrogen production from the first stage AD of FW at different salt concentrations

3.1.1. Hydrogen production rate and cumulative hydrogen yield

As seen in Fig. 1 a, the peak values of average hourly hydrogen production occurred firstly in the salt-free digesters (FW + DW-0 and FW + Air-NBW-0). When salinity was ≥ 20 g NaCl/L, the peak values of average hourly hydrogen production decreased significantly with the increase of salinity under no and addition of Air-NBW ($p = 2.27 \times 10^{-5}$ for the DW group and 1.63×10^{-6} for the Air-NBW group, both $p \ll 0.05$). This indicates that high salinity may inhibit hydrogen production, which is consistent with the negative effects of salt on hydrogen production in previous studies. Kim et al. (2009) obtained 30%, 60%, and 90% inhibition on hydrogen production at 2.41, 5.36, and 10.14 g Na^+/L, respectively. Similarly, Zhang et al. (2017) observed that hydrogen production could be gradually decreased with the salinity increase, and the inhibitory effect was more obviously at salinity ≥ 15 g NaCl/L. It is noteworthy that the peak values of average hourly hydrogen production in the Air-NBW group were higher than those in the DW group under the test salinity conditions (0 – 30 g NaCl/L).

The cumulative hydrogen yield showed similar trends (Fig. 1b) and the maximum hydrogen yield of 197.55 ± 0.63 mL/g-VS_{added} was obtained from FW + Air-NBW-0, about 28% higher than that from FW + DW-0. It is worth mentioning that FW + Air-NBW-5 (185.64 ± 2.53 mL/g-VS_{added}) also obtained approximately 20% higher than FW + DW-0. Moreover, the cumulative hydrogen yields from the Air-NBW added reactors were increased by 30%, 21%, 65%, and 65% in comparison to the DW added reactors at 5, 10, 20, and 30 g NaCl/L, respectively. As seen, salt inhibition on hydrogen production was more obviously in the reactors with DW addition (Fig. 2a). This observation indicates that addition of Air-NBW can effectively ameliorate the inhibition of salt on hydrogen production from the two-stage AD of FW. As reported previously, the inhibition of high concentration salt on AD was mainly caused by the loss of microbial activity (Anwar et al., 2016). However, Air-NBW supplementation can significantly increase the hydrolase activity in the two-stage AD of FW, which may be the reason for the reduced inhibition of salt on hydrogen production under addition of Air-NBW. This would be further discussed in Section 3.1.3.

The kinetic parameters were estimated as shown in Table 2. The maximum hydrogen production potential (P_{H2}) and the maximum hydrogen production rate ($R_{H2,\ max}$) were decreased with the increase of salt concentration, regardless of whether Air-NBW was added or not. However, the values for the reactors with Air-NBW addition were higher than those for the corresponding reactors with DW addition. Upon the increase in salt concentration, the lag time (λ) seemed to increase correspondingly. In contrast, Air-NBW group had about 23 – 42% shorter λ (λ = 6.00 – 14.64 h) in comparison to the DW group (λ = 10.33 – 19.17 h). This observation implies that addition of Air-NBW can accelerate the start-up of the AD process even under salt inhibition conditions.

3.1.2. Potential impact of salt on the hydrolysis/acidification upon Air-NBW addition

The variations of individual VFA, soluble proteins, and soluble

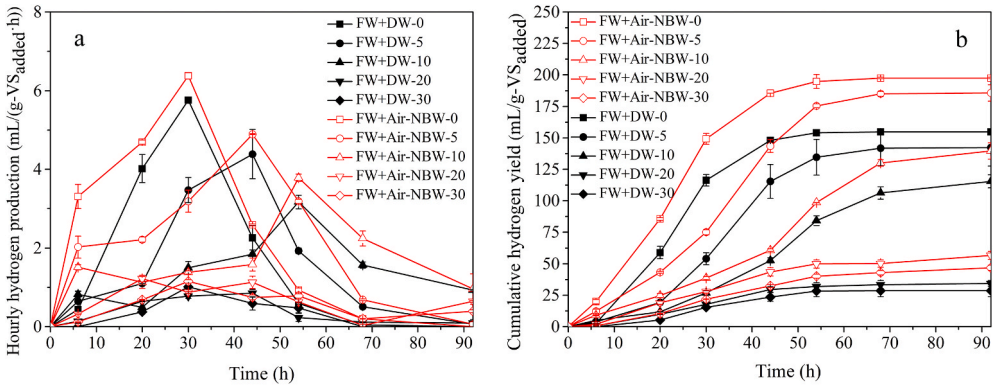


Fig. 1. H_2 production rate (a) and cumulative H_2 yield (b) from FW + DW and FW + Air-NBW at 0 – 30 g NaCl/L.

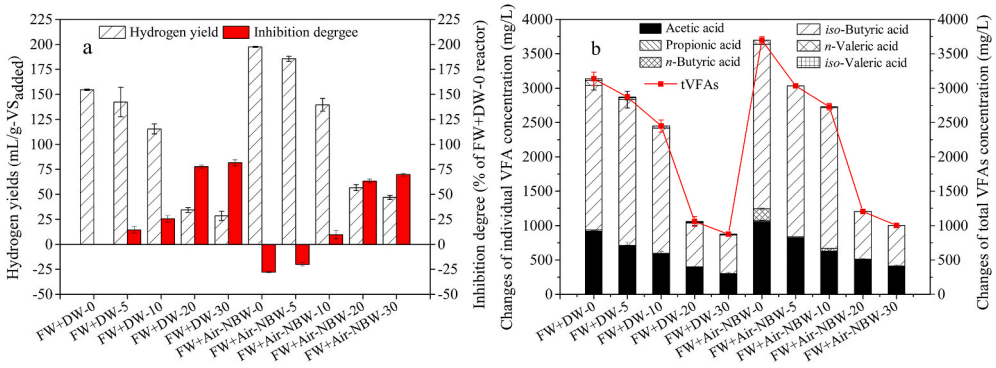


Fig. 2. H_2 yield and inhibition degree (a), and the changes of individual VFA and total VFAs concentration (b) at the end of the first stage AD of FW + DW and FW + Air-NBW at 0 – 30 g NaCl/L.

Table 2
Parameters estimated by fitting the experimental data from the two-stage FW-AD tests with or without Air-NBW addition at different salt concentration using the Modified Gompertz model.

Stage/Parameters	Reactors									
The first-stage AD of FW with or without Air-NBW addition at different salt concentration										
	FW + DW-0	FW + DW-5	FW + DW-10	FW + DW-20	FW + DW-30	FW + Air-NBW-0	FW + Air-NBW-5	FW + Air-NBW-10	FW + Air-NBW-20	FW + Air-NBW-30
P_{H2} (mL/g-VS_{added})	156.49 ± 1.02	146.91 ± 2.86	129.03 ± 6.69	33.75 ± 1.52	28.43 ± 0.90	200.79 ± 3.19	195.46 ± 8.10	156.74 ± 6.61	53.86 ± 2.58	44.61 ± 1.42
$R_{H2,\ max}$ (mL/(g-VS_{added}·h))	17.10 ± 0.15	12.47 ± 0.14	6.13 ± 0.22	4.05 ± 0.51	3.67 ± 0.24	17.85 ± 1.27	13.51 ± 1.68	6.48 ± 0.20	4.86 ± 0.51	3.87 ± 0.17
λ (h)	10.33 ± 0.24	17.16 ± 0.64	17.82 ± 1.25	17.87 ± 1.30	19.17 ± 1.04	6.00 ± 1.18	12.65 ± 2.00	12.88 ± 0.13	13.74 ± 0.54	14.64 ± 1.08
R^2	0.9993	0.9962	0.9901	0.9728	0.9872	0.9965	0.9903	0.9955	0.9737	0.9880
The second-stage AD of FW with or without Air-NBW addition at different salt concentration										
	FW + DW-0	FW + DW-5	FW + DW-10	FW + DW-20	FW + DW-30	FW + Air-NBW-0	FW + Air-NBW-5	FW + Air-NBW-10	FW + Air-NBW-20	FW + Air-NBW-30
P_{CH4} (mL/g-VS_{added})	271.90 ± 2.97	249.21 ± 2.20	231.97 ± 2.25	184.44 ± 0.23	59.87 ± 1.23	323.55 ± 3.38	289.85 ± 2.34	279.17 ± 3.48	208.29 ± 6.23	80.47 ± 0.67
$R_{CH4,\ max}$ (mL/(g-VS_{added}·d))	122.69 ± 1.20	119.36 ± 0.77	90.77 ± 5.65	52.65 ± 5.80	6.76 ± 0.54	138.14 ± 1.67	132.41 ± 2.77	126.28 ± 1.39	55.50 ± 5.73	9.62 ± 0.41
λ (d)	7.52 ± 0.27	8.56 ± 0.11	10.06 ± 0.23	20.71 ± 0.58	22.66 ± 0.25	7.22 ± 0.26	8.52 ± 0.11	9.88 ± 0.29	20.64 ± 0.57	21.23 ± 0.45
R^2	0.9935	0.9959	0.9965	0.9790	0.9870	0.9941	0.9967	0.9935	0.9796	0.9913

carbohydrates may indirectly reflect the degree of hydrolysis/acidification. Therefore, VFAs were monitored in the first stage as shown in Fig. 2b. A remarkably negative effect on VFAs production was noticed with the increase of salt concentrations ($p = 1.19 \times 10^{-6}$ for the DW group and $p = 1.74 \times 10^{-8}$ for the Air-NBW group, both $p \ll 0.05$). It has been found that VFAs yield was inhibited at > 6 g NaCl/L, with the production being reduced at least by 44.8% at > 9 g NaCl/L (Liu et al., 2017). Zhao et al. (2016) claimed that VFAs production was enhanced when salt concentration was < 8 g/L, while severely inhibited when the salt concentration was further increased. In this study, higher total VFAs productions were achieved from the Air-NBW added reactors, increasing by 6 – 18% when compared to the DW added ones. This observation is in agreement with that of hydrogen yield, probably due to the fact that acid production is usually accompanied by hydrogen production.

Regarding the changes of individual VFA concentration, a clear H_2 production through the HAc and HBu pathway was noticed in all the reactors. This indicates that the HBu-type fermentation was the main fermentation type in this study according to Lay (2000) and Zhang et al. (2017). However, at high salinity levels, HAc and HBu productions were reduced, which may be the main reason for the lower H_2 yields in the high salinity reactors.

Ginkel and Logan (2005) detected that a larger amount of the undissociated HAc and HBu was present under lower pH conditions, which led to the inhibited H_2 production. Additionally, the undissociated HBu may exert a more negative effect on H_2 yield than the undissociated HAc. Table 3 summarizes the changes of final pH, undissociated HAc, and undissociated HBu concentrations in all the reactors. As shown, the concentrations of undissociated HAc and HBu were increased with the increase of salinity, while a lower increase was clearly noticed in the Air-NBW added reactors. Heuvel et al. (1992) claimed that the undissociated acid at the critical inhibitory concentration (50 mM) may greatly increase the cell maintenance energy requirements and deplete the ATP reserves necessary for the metabolic switch to solventogenesis. In this study, addition of Air-NBW reduced the accumulation of undissociated acid, which might be resulted from its enhancement effect on the microbial activity as discussed later.

Soluble proteins and carbohydrates play a crucial role in microbial growth and hydrogen production by anaerobic fermentation (Laspidou and Rittmann, 2002). As shown in Fig. 3a and b, the initial concentrations of soluble proteins and carbohydrates were 1014.15 ± 78.82 and 2278.26 ± 128.12 mg/L in all the reactors. After the hydrogen production stage, the concentrations of soluble proteins and carbohydrates declined remarkably in the reactors with lower salinity (0–10 g NaCl/L). However, higher concentrations of soluble proteins and carbohydrates were always detected in the reactors with higher salinity (≥ 20 g NaCl/L), especially in the DW added reactors. Zhao et al. (2017) revealed that the concentrations of soluble carbohydrates and proteins were increased from 8596 and 2156 mg COD/L to 12,054 and 3124 mg COD/L, respectively when NaCl was increased from 2 to 15 g/L in the AD system of FW. This observation suggests that high salinity may accelerate the release of proteins and carbohydrates due to the bacterial protective responses to high salinity stress (Wang et al., 2013). It was noted that the declines in soluble proteins and carbohydrates were more remarkable in the Air-NBW group: (1) they had about 14%, 21%, 48%, 68%, and 113% higher declines in soluble proteins concentrations, and (2) about 11%, 16%, 7%, 13% and 14% higher declines in soluble carbohydrates concentrations than the reactors with DW addition when NaCl varied from 0 to 30 g/L. On the one hand, this phenomenon indicates that the prolonged exposure of microorganisms to high salinity may lead to cell lysis or dissolution, reducing the hydrolysis of proteins and carbohydrates. On the other hand, it also suggests that addition of Air-NBW rather than DW may not only ameliorate the above negative effect but also promote the hydrolysis of organic matters under high salinity conditions. Wang et al. (2017) applied Scanning Electron Microscopy and observed more pores and channels on the surface of sludge after being treated with air micro-bubbles, which is conducive to the formation of zoogloea.

Table 3
The final pH value, undissociated HAc, and undissociated HBu at the end of the first stage AD in each reactor.

Reactors	Final pH	Undissociated HAc (mM)	Undissociated HBu (mM)
FW + DW-0	4.12 ± 0.04	9.94 ± 0.80	17.75 ± 1.22
FW + DW-5	3.84 ± 0.02	14.71 ± 1.46	33.84 ± 0.27
FW + DW-10	3.71 ± 0.06	16.96 ± 1.53	39.78 ± 4.13
FW + DW-20	3.37 ± 0.01	24.76 ± 0.36	30.31 ± 1.58
FW + DW-30	3.14 ± 0.04	32.00 ± 1.62	46.15 ± 3.74
FW + Air-NBW-0	4.36 ± 0.04	6.64 ± 0.49	12.39 ± 1.08
FW + Air-NBW-5	3.96 ± 0.00	12.95 ± 0.15	26.50 ± 0.09
FW + Air-NBW-10	3.85 ± 0.04	12.99 ± 1.35	32.97 ± 2.81
FW + Air-NBW-20	3.58 ± 0.03	19.27 ± 1.06	20.20 ± 1.63
FW + Air-NBW-30	3.39 ± 0.07	25.77 ± 1.23	28.51 ± 0.56

3.1.3. Activities of ETS and extracellular hydrolases during the first stage AD of FW

In this study, the microbial activity indicated by the electron transport rate and extracellular hydrolases was monitored during the first stage. As illustrated in Fig. 3c, the ETS activity was detected to increase and then decrease along with the lower salt concentrations (≤ 10 g NaCl/L); however, in the DW added reactors with 20 and 30 g NaCl/L, a continuous decline of ETS activity was noticed. The highest ETS activity, 71.78 mg/(g·h), was obtained at 44 h in FW + Air-NBW-0. At the end of the first stage, the ETS activities in the Air-NBW added reactors were 5 – 92% higher than the DW added reactors at the test salt concentrations. This observation implies the enhanced electron transfer effect under the Air-NBW addition condition, and thus the increased bioactivity of microbes, which could be ascribed to the involvement of Air-NBs in electron transfer with resultant enhanced synthesis of hydrogen from their negatively charged surfaces. A detailed explanation is given in Section 3.3.

Fig. 3d displays the relative activities of α-glucosidase and protease in all the reactors at the end of the first stage with the extracellular hydrolases of FW + DW-0 being as the reference (100%). Clearly, high salinity may inhibit the activity of hydrolases to some extent when compared to the salt-free reactors. For instance, the relative activity of α-glucosidase was decreased by 13%, 15%, 24%, and 38% in FW + DW-5, FW + DW-10, FW + DW-20, and FW + DW-30, respectively. However, less inhibition was observed on the activity of protease when compared to that of α-glucosidase. For example, the relative activity of protease did not decrease but increased slightly by 2% at salt concentration of 5 g NaCl/L. And the relative activity of protease was decreased by 8%, 17%, and 29% in FW + DW-10, FW + DW-20, and FW + DW-30, respectively. Besides, it is worth noting that 9 – 20% higher α-glucosidase and 0.4 – 22% increase in protease activities were detected in the Air-NBW added reactors than those in the DW added reactors at salt concentrations of 0 – 30 g/L, reflecting the higher bioactivities of anaerobes in the former reactors. Therefore, supplementation of Air-NBW is regarded as an effective strategy to enhance microbial activities under high salinity, even for hydrogen production from the AD of FW.

3.2. Effect of Air-NBW addition on methane production from the second stage AD of FW at different salt concentrations

3.2.1. Methane production rate and cumulative methane yield

In the second stage, the remaining hydrolyzed and acidified organic slurries at the end of the first stage were used as the substrates for methane production. Much attention was paid to the effect of salt on methane production from the second stage AD of FW with or without Air-NBW addition. The daily methane production as shown in Fig. 4a clearly indicates some prolongation effect of salt on the reactor start-up and progress. More specifically, when the salt concentration was 20 g NaCl/L (FW + DW-20), the peak daily methane production was reached on day 30, with 18 days later than FW + DW-0; furthermore, when the salt concentration was 30 g NaCl/L (FW + DW-30), the reactor failed to reach the peak daily methane production till the end of the experiment (day 36). There are some halophilic anaerobic microbial communities (Lefebvre et al., 2006), while these halophilic microorganisms are

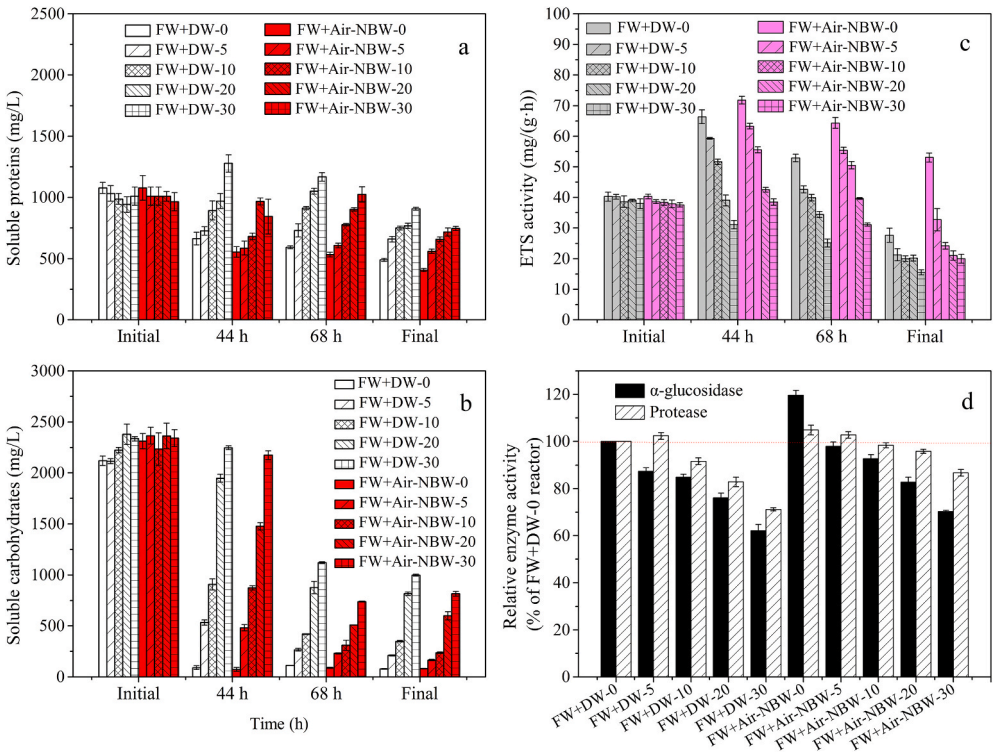


Fig. 3. The concentrations of soluble proteins (a) and carbohydrates (b) during the first stage AD of FW + DW and FW + Air-NBW, and the effects of salt on ETS activity (c) and relative activities of α-glucosidase and protease (d) at the end of the first stage AD of high salinity FW with or without Air-NBW addition.

limited, and non-halophilic microorganisms require a long time to accommodate the high salinity condition. Therefore, when coping with high salinity FW, anaerobic microbes should be given enough time to adapt to the high salinity environment, thus a longer start-up time was required at high salinity (like 20 and 30 g NaCl/L in this study). The lag time (λ) estimated by the Modified Gompertz model shows a similar result (Table 2).

Regarding the cumulative methane yield (Fig. 4b), a significant reduction was observed at high salinity, especially in the DW added reactors. In the Air-NBW added reactors, the cumulative methane yields were increased by 19%, 16%, 20%, 14%, and 43% when compared to the DW added reactors at 0, 5, 10, 20, and 30 g NaCl/L, respectively. As seen from Fig. 4c, the inhibition degrees of methane production were 8%, 14%, 34%, and 83% in FW + DW-5, FW + DW-10, FW + DW-20, and FW + DW-30, respectively. Compared with previous studies, the two-stage AD of high-salt FW in this study was more resistant to salt inhibition on methane production. For instance, Anwar et al. (2016) claimed that the inhibition degrees of one-stage kitchen waste AD system were 24%, 33%, 56%, and 81% in the reactors at 10, 12, 14, and 16 g NaCl/L, respectively. And 80% and 91% of inhibition degrees were obtained on methane production from the one-stage AD of FW at 10 and 15 g NaCl/L, respectively (Zhao et al., 2017). Two-stage AD of FW used in this study may add some mitigation effect of salt inhibition on methane production because the microorganisms in the first stage have been exposed and got accommodated to the high salinity conditions. Additionally, it is interesting to notice that the methane yields from the Air-NBW added reactors at 0, 5, and 10 g NaCl/L were increased by about 19%, 7%, and 3% in comparison to that from FW + DW-0, and the inhibition degrees of methane production were only 26% and 76% in FW + Air-NBW-20 and FW + Air-NBW-30, respectively. These results imply that addition of Air-NBW could also effectively reduce the salt inhibition on methane production in the second stage AD of FW.

3.2.2. *Variations of organic matters concentration during the second stage AD*

The initial and final concentrations of soluble proteins and carbohydrates in the second stage AD are displayed in Fig. 5a. The initial concentrations of soluble proteins in all the reactors were higher than those of soluble carbohydrates. It is probably because the hydrolysis of proteins in the first stage was not as sufficient as that of carbohydrates, resulting in more residual proteins. The hydrolysis of proteins has been reported later than that of carbohydrates and thus required a longer hydrolysis time (Kim et al., 2019), which was also verified in this study. Especially under high salt concentrations, the initial concentrations of soluble proteins and carbohydrates in the second stage were higher, which need to be further hydrolyzed in the second stage with a prolonged lag time. This could also be one of the reasons for the prolonged start-up of methane production with the increase of salt concentration.

The VFAs produced in the first stage were the primary substrates that could be directly converted into methane by methanogens during the second stage. Therefore, a high initial VFAs concentration in the second stage is conducive to methane production (De Gioannis et al., 2017). As shown in Fig. 5b, higher initial VFAs concentrations were detected in the reactors with low salt concentrations (0 – 10 g NaCl/L), and addition of Air-NBW could produce more VFAs in the first stage, which can be utilized by methanogens in the second stage. However, when the salt concentration was 30 g NaCl/L, the total VFAs concentration was not decreased but increased at the end of the methane production stage, particularly in FW + DW-30. Due to the hydrogen production metabolic pathway, the first stage was dominated by HBu, which was followed by HAc with little propionic acid (HPr) production (the initial VFAs for the second stage). Nevertheless, HPr was accumulated, around 180 and 285 mg/L in the FW + DW-20 and FW + DW-30 at the end of the second stage, respectively. HAc is regarded as the preferable substrate for methane production, while HPr is not conducive to the utilization by methanogens (Capson-Tojo et al., 2018). These data suggest that the inhibition of high salinity (≥ 20 g NaCl/L) on methanogenesis may be attributable to low efficiency of hydrolysis/acidification in the first stage

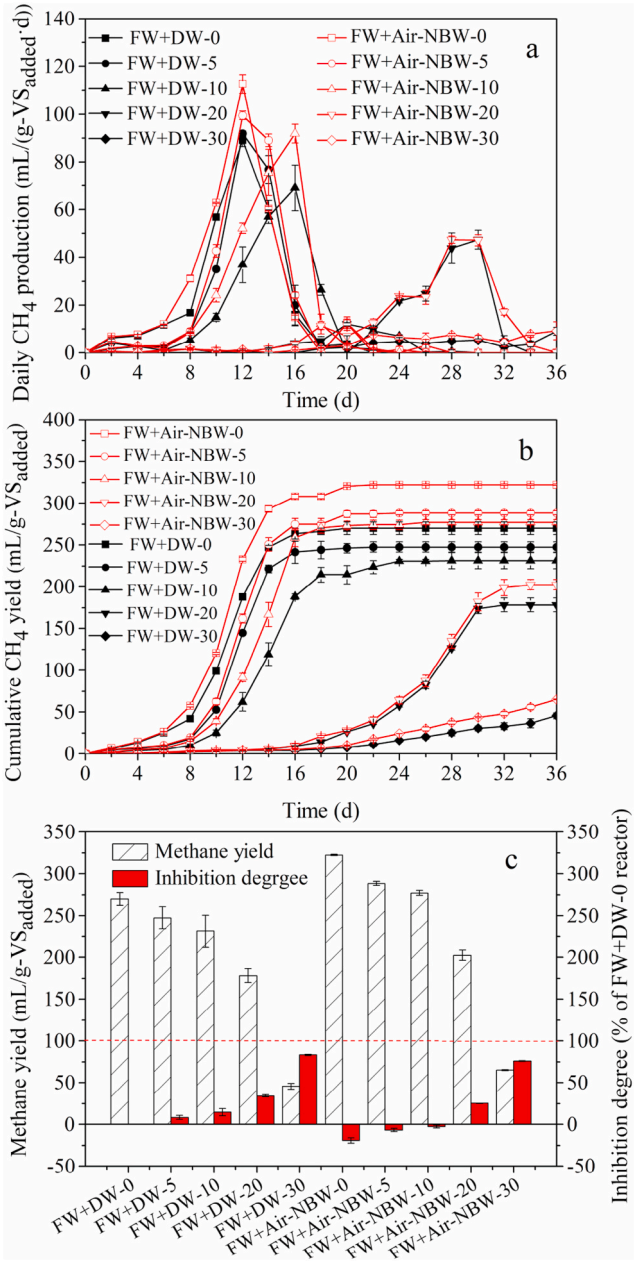


Fig. 4. CH_4 production rate (a), cumulative CH_4 yield (b), and CH_4 yield and inhibition degree (c) during the second stage AD of FW + DW and FW + Air-NBW at 0 – 30 g NaCl/L.

and unfavorable methanogenesis due to the produced HPr.

3.2.3. Activities of ETS and coenzyme F_{420} in the second stage AD of FW at different salt concentrations

The ETS activities in the reactors with or without Air-NBW addition were assessed during the 36 days' methanogenesis as shown in Fig. 5c. In the reactors with low salinity (0 – 10 g NaCl/L), the significant increase in ETS activity was detected, especially in the Air-NBW added reactors. However, the ETS activities in FW + DW-20 and FW + DW-30 were decreased slightly during the first 12 days, reflecting the negative effect of high salinity on ETS of microorganisms and thus reduced microbial activity. On the contrary, the ETS activities in the FW + Air-NBW-20 and FW + Air-NBW-30 were increased continually during the whole process, indicating that addition of Air-NBW was beneficial for mitigating the adverse effect of high salinity on ETS activity in the methanogenesis. At the end of the second stage, the ETS activity was increased by 9 – 30% in the Air-NBW added reactors when compared to the DW added reactors. In addition, after the 36 days' methane production, the relative activities of coenzyme F_{420} in all the reactors exhibited the same trend (Fig. 5d), i.e., when FW + DW-0 was taken as the reference (100%), the activity of coenzyme F_{420} was detected to remarkably decrease at high salinity levels, which were 73%, 62%, 54%, and 40% in the DW added reactors, and 130%, 110%, 99%, 72%, and 49% in the Air-NBW added ones at 5, 10, 20, and 30 g NaCl/L, respectively. These results indicate that addition of Air-NBW could enhance methane production by increasing microbial activity even under high salinity conditions.

3.3. Preliminary analysis of mechanisms involved in this study

Salt toxicity to microorganisms is predominantly determined by cations (Anwar et al., 2016), and the main cation in FW is Na^+ because FW salinity mainly derives from food flavoring agents (NaCl). Osmotic pressure would increase sharply at high salt concentrations leading to plasmolysis and ultimately cell lysis and death (Fig. 6). This is why high salinity has a negative effect on AD. Meegoda et al. (2019) found that negatively charged NBs in an electrolyte solution (0.001 – 0.1 M NaCl) could combine with Na^+ to form the electric double layer, which may prevent Na^+ from entering the cell, thus reducing adverse effects of Na^+ on microorganisms. However, the effect of high salinity on the morphology of NBs cannot be ignored. As reported by Das (2011) and Bunkin and Shkirin (2012), low salinity exerts an imperceptible effect on physicochemical characteristics of NBs, but when the NBs are exposed to a high salinity solution for a while, the bubble size, surface charge density, and the number of negative charges would increase. Nevertheless, the possible effect of these observations on AD is unclear.

As shown in Fig. 6, during the H_2 production stage, addition of Air-NBW enhanced the activities of ETS and extracellular hydrolases at different salt concentrations, which are likely attributable to the involvement of Air-NBs in the electron transfer associated with the synthesis of hydrogen and enhanced biosynthesis of hydrolases. More specifically, the negative charges on the surfaces of NBs could attract H^+, which may be more easily reduced by NADH to H_2 following $NADH+H^+ \rightarrow NAD^+ + H_2$ (Cai et al., 2011). Tanisho et al. (1989) also found that the conversion of NADH to hydrogen could occur on the cell membranes, implying that the hydrogenase involved in the NADH pathway is a membrane-bound enzyme. Therefore, Air-NBs may promote the electron exchange between NADH ferredoxin oxidoreductase (NFOR) and hydrogenase to produce H_2 due to it is negatively charged. For the methanogenesis stage, addition of Air-NBW also improves the activities of ETS and coenzyme F_{420}, most probably attributable to its stimulation effect on electron transfer between bacteria and methanogens. It also indirectly indicates that addition of Air-NBW enhanced microbial activity, resulting in increased CH_4 production. Liu et al. (2015) and Xiao and Xu (2020) have verified that NBW could accelerate the metabolism of organisms, facilitating seed germination and biofilm growth. Still, more direct evidences are demanding for the above viewpoints in the followed-up works.

4. Conclusions

In this study, the hydrolysis/acidogenesis and methanogenesis stages were seriously inhibited at high salt levels (≥ 20 g NaCl/L). However, addition of Air-NBW reduced the inhibition of high salinity on the efficiency of two-stage AD of FW, with hydrogen and methane yields increased by 21 – 65% and 14 – 43% in the Air-NBW added reactors in comparison to the DW added ones at 0 – 30 g NaCl/L. α-glucosidase and protease activities were enhanced by 9 – 20% and 0.4 – 22% at the end of the first stage, and coenzyme F_{420} activity was increased by 26 – 58% at the end of the second stage in the Air-NBW group with 0 – 30 g NaCl/L, respectively. Certainly, future works are necessary to emphasize the NBW application and performance in the large-scale and long-term

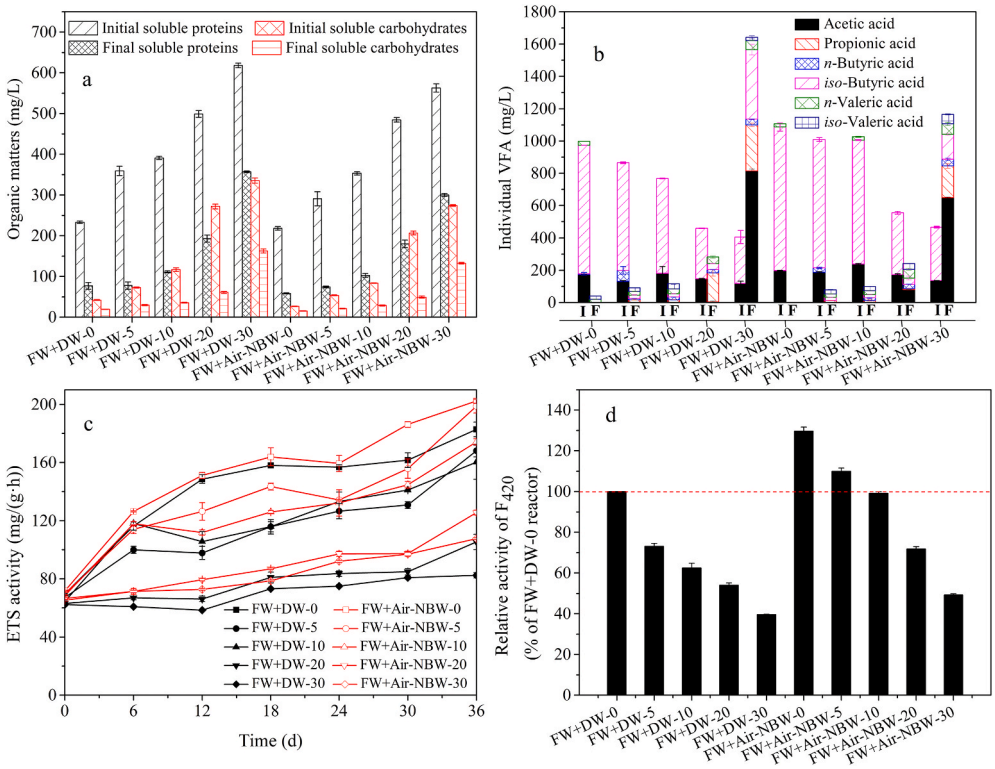


Fig. 5. The concentrations of initial and final organic matters (a) and individual VFA (b) during the second stage AD of FW + DW and FW + Air-NBW at 0 – 30 g NaCl/L; the effects of salt on ETS activity (c) during the methanogenesis, and the relative activities of coenzyme F_{420} (d) at the end of the second stage AD of high salinity FW with or without Air-NBW addition. The letters 'I' and 'F'at X-axis denote the initial and final concentrations, respectively.

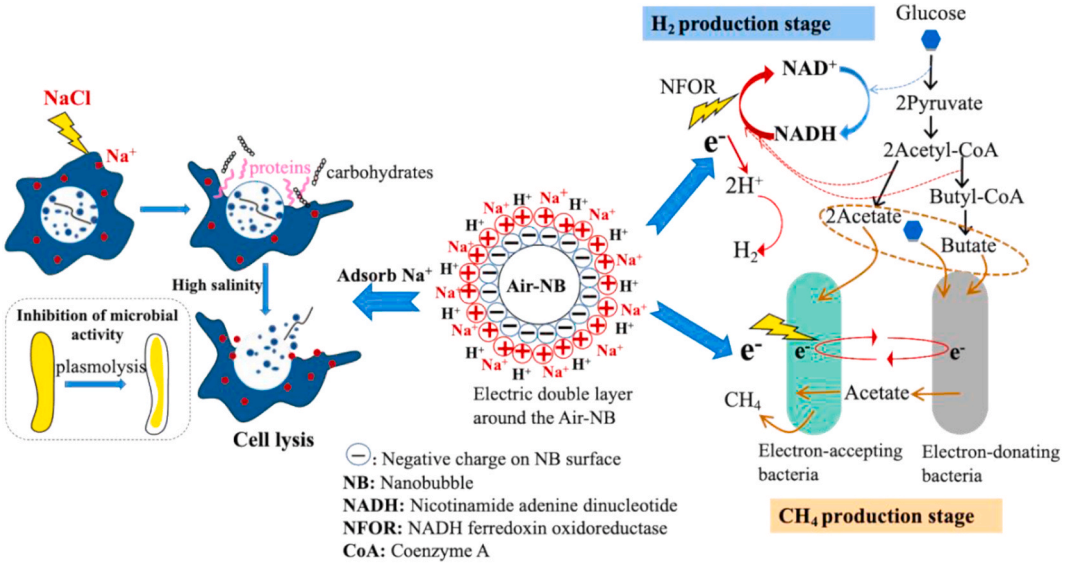


Fig. 6. Impact of NaCl on microorganisms and preliminary analysis of mechanisms involved in the enhanced H_2 and CH_4 production from two-stage AD of high salinity FW with Air-NBW addition and resultant promoted electron transfer.

operation of two-stage AD of FW systems.

CRediT authorship contribution statement

Tingting Hou: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Writing – original draft, Writing – review & editing. **Jiamin Zhao:** Data curation. **Zhongfang Lei:** Conceptualization, Formal analysis, Funding acquisition, Supervision, Writing – review & editing. **Kazuya Shimizu:** Formal analysis, Methodology, Supervision. **Zhenya Zhang:** Conceptualization, Project administration, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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